

SHERBROOKE LENNOXVILLE, QC, Canada

WMO: 716110

Lat: 45.3686N

Lon: 71.8233W

Elev: 594

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -13.7	(c) -8.1	(d) -20.9	(e) 1.8	(f) -12.8	(g) -15.8	(h) 2.4	(i) -7.0	(j) 23.3	(k) 25.6	(l) 20.9	(m) 23.6	(n) 4.1	(o) 60	(p) 0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.5	(c) 85.0	(d) 70.7	(e) 82.3	(f) 69.1	(g) 79.8	(h) 67.8	(i) 73.5	(j) 81.1	(k) 71.7	(l) 79.0	(m) 70.0	(n) 76.8	(o) 9.5	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	117.3	77.5	69.3	110.0	75.7	67.6	103.6	73.8	37.5	81.4	35.9	79.1	34.4	76.9	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	17.5	15.4	-23.3	89.6	5.7	1.8	-27.4	90.9	-30.8	91.9	-34.0	92.9	-38.2	94.2
			-22.9	76.3	6.4	2.0	-27.5	77.8	-31.2	79.0	-34.8	80.1	-39.5	81.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.3	15.6	18.4	27.8	41.5	54.3	63.0	67.5	66.1	58.8	46.5	35.1	23.0
	DBStd	20.36	13.10	11.74	11.39	8.49	7.67	6.56	5.21	5.20	7.34	8.27	9.61	11.34
	HDD50	4436	1065	885	692	279	41	1	0	0	11	172	453	837
	HDD65	8216	1530	1305	1155	704	340	114	32	51	211	576	896	1302
	CDD50	1981	0	0	2	25	176	391	543	499	276	62	7	0
	CDD65	285	0	0	0	0	10	53	110	85	26	1	0	0
	CDH74	2497	0	0	1	17	161	521	897	654	236	10	0	0
	CDH80	512	0	0	0	2	28	122	197	117	46	0	0	0
(14) Wind	WSAvg	6.4	6.8	6.9	7.3	7.5	6.8	5.8	5.3	4.9	5.2	6.4	6.7	6.7
(15) Precipitation	PrecAvg	43.1	2.8	2.5	2.8	3.2	3.8	4.0	4.4	4.7	3.5	3.3	3.8	3.7
	PrecMax	55.8	7.6	5.2	4.5	7.2	6.5	6.5	7.7	8.1	6.6	8.0	6.5	7.4
	PrecMin	34.8	0.7	0.4	0.5	1.4	1.5	1.5	2.1	1.7	1.5	0.9	2.1	1.7
	PrecStd	5.4	1.4	1.2	1.0	1.3	1.6	1.3	1.7	1.5	1.3	1.5	1.1	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.0	50.2	63.2	76.1	83.8	87.5	88.7	86.8	85.4	75.3	66.0	51.2
		MCWB	46.8	44.6	51.7	59.3	65.7	71.1	74.4	72.5	71.7	63.5	58.6	47.6
	2%	DB	42.4	43.2	53.3	69.0	78.5	83.6	84.9	83.2	80.0	69.2	59.2	46.2
		MCWB	39.7	38.4	44.7	54.4	61.7	69.1	71.3	70.2	68.4	59.6	52.9	43.9
	5%	DB	38.1	39.1	47.7	62.6	74.1	80.1	81.8	80.5	76.1	65.0	53.9	41.4
		MCWB	35.9	35.4	41.0	50.6	60.2	67.1	69.5	69.0	66.1	57.7	48.3	39.3
	10%	DB	34.2	35.8	43.1	56.6	69.8	76.4	79.2	77.7	72.2	60.4	49.1	37.6
		MCWB	32.3	33.0	38.3	46.9	58.1	65.3	68.0	67.5	64.6	54.2	45.2	35.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.7	46.0	55.5	62.9	69.7	74.4	76.9	74.8	73.0	66.2	59.7	48.4
		MCDB	48.2	48.9	61.8	72.8	78.4	83.0	85.8	82.5	81.9	72.7	64.1	50.7
	2%	WB	40.1	39.5	46.1	56.9	65.9	71.9	73.8	72.7	70.5	61.7	54.3	44.0
		MCDB	41.8	42.4	50.7	65.5	73.9	80.0	81.0	79.9	77.3	66.9	58.7	45.9
	5%	WB	35.9	35.8	41.9	51.9	62.9	69.4	71.9	70.8	68.2	58.0	49.4	39.5
		MCDB	38.2	38.6	46.4	58.5	70.5	76.6	78.9	77.3	74.2	63.6	52.8	41.3
	10%	WB	32.4	33.2	38.7	48.0	59.7	67.1	70.0	69.1	65.7	55.0	45.2	35.8
		MCDB	34.2	35.7	42.9	55.8	67.1	73.8	76.8	75.3	70.9	59.9	48.7	37.3
(35) Mean Daily Temperature Range	5% DB	MDBR	17.9	19.7	18.7	18.9	21.0	19.9	19.5	20.0	20.7	17.3	14.5	14.5
		MCDBR	18.9	21.0	24.8	30.5	28.7	25.5	22.6	23.7	25.4	24.8	22.3	15.7
	5% WB	MCWBR	17.6	16.7	16.7	18.0	14.9	12.9	11.3	12.6	14.5	15.9	16.7	14.8
		MCDWR	18.5	18.6	21.2	25.6	23.2	21.2	19.6	20.0	21.9	21.8	20.1	15.2
(40) Clear-Sky Solar Irradiance	taub		0.265	0.285	0.315	0.350	0.374	0.416	0.422	0.401	0.368	0.329	0.306	0.277
		taud	2.372	2.282	2.333	2.382	2.379	2.254	2.245	2.306	2.397	2.505	2.490	2.390
	Ebn at Noon		271	285	291	289	284	271	267	269	269	267	254	253
		Edh at Noon	23	31	34	36	37	42	42	38	32	25	21	21
(44) All-Sky Solar Radiation	RadAvg		445	728	1111	1378	1563	1731	1731	1539	1243	719	437	312
	RadStd		60	88	84	155	173	133	145	89	103	69	57	30

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	N/A

Nomenclature: See separate page